


Vector calculus ^{mod.} exam.

$$1. (a) \nabla \times B = \begin{pmatrix} \partial_y B_z - \partial_z B_y \\ \partial_z B_x - \partial_x B_z \\ \partial_x B_y - \partial_y B_x \end{pmatrix} = \begin{pmatrix} ay - 0 \\ 0 - 0 \\ \cos y - cx \cos y \end{pmatrix}$$

$$\begin{cases} ay = ay \\ bx^2 = 0 \\ (1-x) \cos y = (1-cx) \cos y \end{cases} \Rightarrow \begin{cases} a=1 \\ b=0 \\ c=1 \end{cases}$$

$$\nabla \cdot A = 0$$

(b) any of a & b

$$1(c) (i) \nabla \times F = \begin{pmatrix} 0 - 0 \\ 2z \sinh x + \beta z \sinh x \\ -2 \sinh x - \alpha \sinh x \end{pmatrix}$$

$$\begin{cases} \beta + 2 = 0 \\ \alpha + 2 = 0 \end{cases} \Rightarrow \alpha = \beta = -2$$

$$(ii) \nabla \phi = (- (ry - z^2) \sinh x, r \cosh x, -2z \cosh x)$$

$$\begin{cases} r = -\alpha \\ r = 2 \\ \beta = -2 \end{cases} \Rightarrow \begin{cases} \alpha = -2 \\ \beta = -2 \\ r = 2 \end{cases}$$

$$(iii) \int_c F dr = F(r(1)) - F(r(0)) = -4$$

$$2. \quad (a) \quad F = \nabla f = (2x, 2y, 1)$$

$$\nabla \times F = \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ x & y & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \vec{0}$$

$$\nabla \cdot F = 4$$

$$(b) \quad f = \rho^2 + z \quad F = (2\rho \cos \phi, 2\rho \sin \phi, 1)$$

$$(c) \quad \nabla f = (2\rho, 0, 1)$$

$$\nabla^2 f = 4$$

$$3. \quad (a) \quad \int_S \nabla \times A \cdot \vec{n} \, da = \int_C A \cdot d\vec{l}$$

$$(b) \quad \int_C \vec{r} \cdot d\vec{r} = \int \nabla \times \vec{r} \cdot \vec{n} \, da = 0$$

$$\begin{aligned} (c) \quad I &= \oint_C (y, -x, 0) \, d\vec{r} \\ &= \int_0^{2\pi} (\sin t, -\cos t, 0) \cdot (-\sin t, \cos t, 0) \, dt \\ &= -2\pi \end{aligned}$$

$$\begin{aligned} (d) \quad I_z &= \iint_S \vec{G} \cdot \vec{n} \, dS = \iint_S (x, y, -2-2z) \cdot (x, y, z) \, dS \\ &= \iint_S x^2 + y^2 - 2z - 2z^2 \end{aligned}$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\pi} x (1 - \cos\theta - 3\cos^3\theta) \sin\theta \, d\theta \, d\phi$$

$$= -2\pi$$

(e) $G = \nabla \times A$

Electromagnetism mode paper-

4. (a) $\nabla \times E = 0$

$$E = -\nabla V$$

$$V = - \int_0^1 E(r(t)) \frac{dr}{dt} \cdot dt$$

$$= \alpha e^{-\rho r^2}$$

1b) (ii) $\iint_{\partial V} E \cdot d\vec{a} = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{2L\lambda}{\epsilon}$

(iii) $E \cdot 2\pi R \lambda L = \frac{\lambda L}{\epsilon}$

$$E = \frac{\lambda}{2\pi R \epsilon}$$

5. (a) $\frac{\rho}{\epsilon} = \nabla \cdot E = 0$

1b) $\nabla \times E = -\frac{\partial}{\partial t} B$

$$\nabla \times E = \dots$$

Mode exam.

